EXPERIMENT 4

HASH COMPARISON USING HASHCALC

Date: 07/01/2026

AIM

Using **Hash Calc** write down the Hash values of popular hashing algorithms and do Hash comparison of different files

THEORY:

HashCalc is an easy-to-use tool that allows to compute message digests, checksums and HMACs for files, as well as for text and hex strings. It offers a choice of 12 most popular hash and checksum algorithms for calculations.

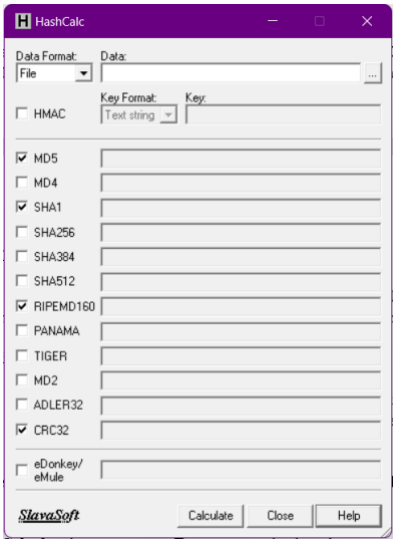
Major features**:**

- Support of 12 well-known and documented hash and checksum algorithms: MD2, MD4, MD5, SHA1, SHA256, SHA384, SHA512, RIPEMD160, PANAMA, TIGER, ADLER32, CRC32;
- Support of the MD4-based hash algorithm used in many P2P applications (edonkey, emule, etc.);
- Support of 2 modes of calculations: HASH/CHECKSUM and HMAC;
- Support of 3 input data formats: files, text strings, and hex strings;
- Work with larger files also

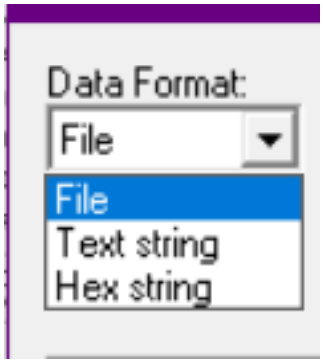
ALGORITHM

Step 1: Install Hash Calc **https://hashcalc.en.softonic.com/**

Step 2: Run HashCalc. A window gets opened when you run it.



Step 3: In the **Data Format** box, select the type of data you want to use for calculations. HashCalc allows to perform calculations on files, text strings or hex strings.

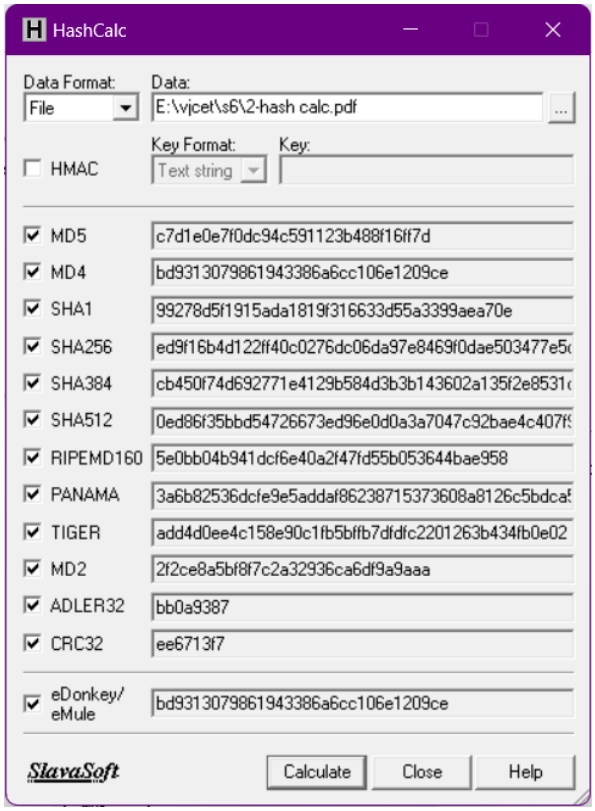


Step 4: In the **Data** box enter the data for calculations:

Step 4.1: It is a file name if 'File' Data Format is selected. Type the file name in the **Data** box, or use the **Find File** button to select the file, or drag and drop the file over the **HashCalc** window.

Step 4.2: it is a string of characters if 'Text string' Data Format is selected.

Step 4.3: it is a string of hexadecimal digits if 'Hex string' Data Format is selected.



Step 5: To calculate the message digests / checksums for the data, the **HMAC** box must be unchecked. To calculate the HMAC (Keyed - **H**ash **M**essage **A**uthentication **C**ode) for the data

Step 5.1: Check the **HMAC** box.

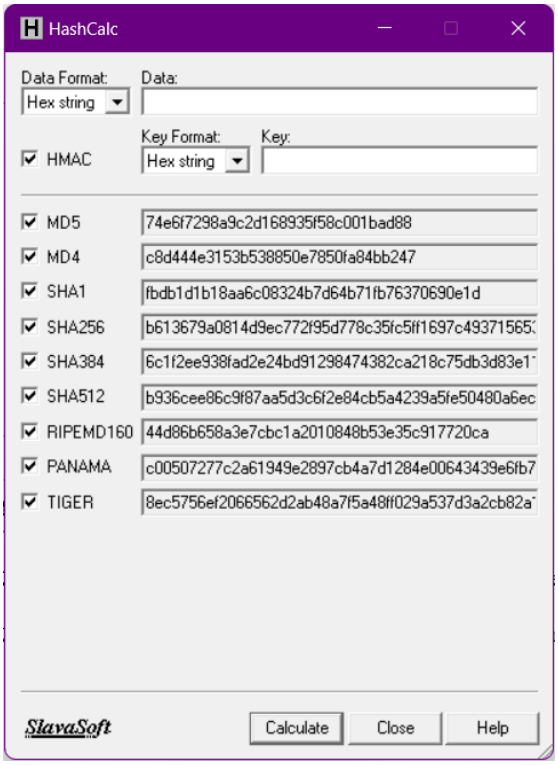
Step 5.2: In the **Key Format** combo box, select the type of the key you want to use for calculations. HashCalc allows to perform calculations using text keys or hex keys.

Step 5.3: In the **Key** box enter the key for HMAC calculations:

- It is a string of characters if 'Text string' Key Format is selected.
- Or it is a string of hexadecimal digits if 'Hex string' Key Format is selected.

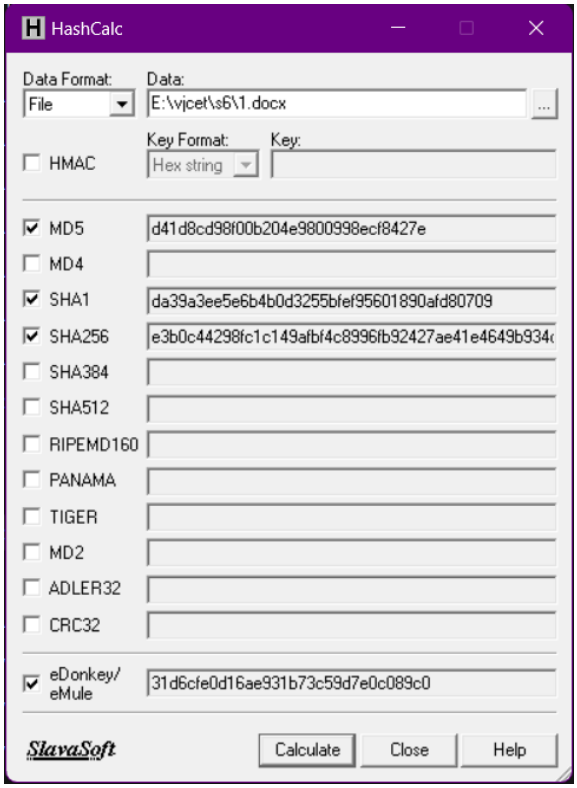
Step 6: Select the algorithms you want to use for calculations by checking the boxes with appropriate names.

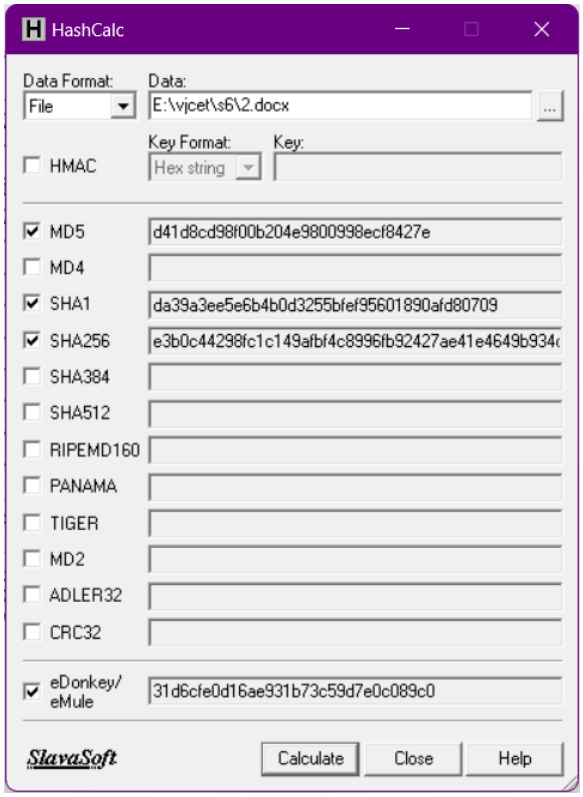
Step 7: Click the **Calculate** button, or press **Enter** on your keyboard.



OUTPUT

Find MDS and SHA1 of two blank files 1.docx and 2.docx and compare their hash values When we calculated the hash values of two blank files, we obtained same hash values.So the integrity of the file is preserved.





Result: Analysed the Hash values using different hashing algorithms and the output was verified successfully.